

BCSE497J - Project-I

**REGION-GROUNDED FORCED-CHOICE ATTRIBUTE
VERIFICATION FOR MULTIMODAL HALLUCINATION
DETECTION**

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ABSTRACT

Vision–language models can generate fluent descriptions while assigning attributes to objects that are not supported by an image. This project formulates attribute hallucination detection as a visual verification problem and studies two commonly proposed remedies: grounding the relevant object region and replacing absolute thresholding with comparative forced-choice decision making. The implementation uses OWLv2 for open-vocabulary object localisation and SigLIP for image–text scoring, with no large language model or paid API. A controlled 2×2 experiment evaluates whole-image versus cropped-region inputs and threshold-based versus forced-choice decisions on the AMBER benchmark. On the held-out test set, whole-image thresholding obtains 57.51% accuracy, whole-image forced choice 77.87%, cropped-region thresholding 59.53%, and cropped-region forced choice 80.12%. Thus, the combined method improves over the baseline by 22.60 percentage points. The ablation attributes +20.36 points to forced choice and only +2.01 points to region grounding. The project therefore shows that changing the decision formulation is substantially more important than localisation for this attribute-verification setting. The system uses two open-weight models totaling 358 million parameters and peaks at 1.65 GiB of VRAM, making it suitable for offline execution on a student laptop. The study also documents benchmark construction artifacts and uses held-out evaluation to reduce the risk of overstating the observed effect.

Keywords—Multimodal hallucination, vision–language models, attribute verification, region grounding, forced choice, SigLIP, OWLv2, AMBER.

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1. INTRODUCTION

1.1 Background

Vision-Language Models (VLMs) and Multimodal Large Language Models (MLLMs) jointly process visual and textual information and are used for image captioning, visual question answering, image understanding, and multimodal assistants. Their fluent outputs, however, do not guarantee that every statement is grounded in the input image. A particularly difficult failure occurs when a model identifies the correct object but assigns it an unsupported attribute, such as an incorrect colour, state, size, action, or material [1], [2].

Attribute hallucination differs from simple object hallucination. If an object is absent, an existence detector can often identify the error by checking whether the object is present. In attribute hallucination, the object may be genuinely visible and correctly named; the error lies in the property attached to that object. This makes the task closer to fine-grained visual verification than to object detection.

1.2 Motivation

Recent hallucination benchmarks and verification pipelines have established that generated claims can be decomposed and checked against evidence. However, many systems still score a claim against the whole image and convert an absolute similarity score into a decision using a threshold. Contrastive vision-language models can struggle with attribute–object binding, and an absolute score has no universal meaning across objects, images, lighting conditions, and prompts [1], [2].

Two intuitive changes are therefore considered in this project. First, region grounding can restrict visual evidence to the object named by the claim. Second, forced-choice verification can replace the question “Is this score high enough?” with “Which of these two competing attributes scores higher?” The project is motivated by the need to measure these two ingredients separately rather than combining them and reporting only a single final number.

1.3 Scope of the Project

The project focuses on English attribute-based verification using the AMBER benchmark. The implementation concentrates on state and action questions for which paired positive and negative attributes can be identified. OWLv2 provides zero-shot open-vocabulary localisation, while SigLIP performs image–text scoring. No model training or fine-tuning is performed. The evaluation is limited to AMBER’s closed object vocabulary and to the controlled whole-image/cropped-region and threshold/forced-choice comparison.

2. PROJECT DESCRIPTION AND GOALS

2.1 Literature Review

The literature is divided into two groups. SemEval work provides the broader hallucination-detection context, while multimodal hallucination benchmarks and verification methods provide the direct technical background for visual attribute verification.

2.1.1 SemEval / SHROOM Literature

Paper / Task	Objective	Methodology	Relevance / Limitation
SHROOM — SemEval-2024 [4]	Detect hallucination in generated text.	Binary hallucination detection with model-agnostic and model-aware tracks and human labels.	Establishes hallucination detection as a measurable verification task, but is text-only.
SHROOM-INDElab	Improve SHROOM classification.	Zero-/few-shot LLM prompting with task-specific prompts.	Shows prompt-based strategies can work without task-specific training, but remain model/prompt dependent.
MALTO	Improve SHROOM detection with synthetic data.	LLM pseudo-labeling, rephrasing, and NLI ensemble.	Demonstrates the value of augmentation, but depends on synthetic labels.
Mu-SHROOM — SemEval-2025 [5]	Extend hallucination detection to multiple languages.	Multilingual span-level identification of hallucinated content across 14 languages.	Extends systematic hallucination evaluation, but still focuses on generated text rather than visual attribute grounding.

Table 1. SemEval literature relevant to the adopted problem formulation.

2.1.2 Multimodal Hallucination and Dataset Literature

Work / Benchmark	Focus	Dataset / Evaluation	Relevance to this project
Winoground [1]	Visio-linguistic compositionality.	Paired images and captions with changed meanings.	Shows that models can recognise objects but still fail at relationships and compositional binding.
ARO [2]	Attributes, relations, and word order.	Attribute–object binding and related compositional tests.	Direct evidence that attribute binding is difficult for vision-language models.
AMBER [3]	Existence, attribute, and relation hallucination.	1,004 images; 14,216 discriminative queries plus generative evaluation.	Primary benchmark; directly provides attribute questions and human-verified labels.
FaithScore [6]	Fine-grained verification of generated responses.	Atomic fact decomposition and visual verification.	Supports claim-level verification but does not isolate the attribute decision rule.
Woodpecker [7]	Hallucination detection and correction.	Concept extraction, question generation, and visual verification.	Shows claims can be checked against visual evidence, but is a broader pipeline.
UNIHD [8]	Unified multimodal hallucination detection.	Specialised tools for different hallucination categories.	Highlights that attribute hallucination remains

Work / Benchmark	Focus	Dataset / Evaluation	Relevance to this project
			challenging and does not isolate decision strategy.
ESREAL [9]	Region-level visual hallucination analysis.	Relevant-region reconstruction / analysis.	Motivates region-level evidence, but does not separately measure localisation versus decision strategy.
POPE [10]	Object-existence hallucination.	MS-COCO binary yes/no probing.	Demonstrates the value of discriminative evaluation, but is primarily object existence.
HallusionBench [11]	Visual-language reasoning hallucinations.	Paired visual reasoning cases.	Covers broader visual reasoning failures beyond the attribute setting.
PhD [12]	Fine-grained visual hallucination evaluation.	Large-scale daily and counter-common-sense images with VQA triplets.	Shows movement toward task-specific visual hallucination benchmarks.

Table 2. Multimodal hallucination and dataset literature considered in the project.

Dataset used by the system. AMBER is the primary benchmark. The project uses its discriminative state and action questions, where a true and false attribute are associated with the same image and object. From the relevant questions, 2,378 clean state pairs and 396 action pairs were verified for the forced-choice evaluation path. This paired structure supplies the two competing attributes without requiring an independently constructed antonym lexicon.

2.2 Research Gap

- Existing research shows that vision-language models have difficulty understanding attributes correctly and connecting them to the right object.
- General hallucination detection and verification systems can check generated claims against images, but many combine several techniques in a single pipeline.
- Region-level visual information has been proposed to focus verification on relevant image evidence, yet its independent contribution is difficult to quantify.
- Threshold-based verification asks whether an individual score exceeds a cutoff, whereas comparative forced choice asks which competing attribute scores higher.
- Most existing systems do not isolate these two factors under the same data, prompts, model, and evaluation protocol.

Therefore, this project performs a controlled 2×2 experiment that independently varies visual input (whole image versus cropped region) and decision rule (threshold versus forced choice). The central research question is: Which contributes more to attribute verification—region grounding or comparative forced-choice decision making?

2.3 Objectives

1. To develop a system for verifying attribute-based claims against images using lightweight open-source vision models.

2. To study whether using the relevant object region improves attribute verification compared with using the complete image.
3. To compare threshold-based verification with forced choice between competing attributes.
4. To separately measure the effect of region grounding and forced choice using a controlled 2×2 experiment.
5. To evaluate the proposed method on the AMBER benchmark using a held-out test set.

2.4 Problem Statement

Given an image and a natural-language attribute claim, determine whether the claim is visually supported by the relevant evidence. The system should identify the object referred to by the claim, obtain visual evidence, score competing attribute descriptions, and produce a binary supported/unsupported decision. The evaluation must distinguish the effect of region grounding from the effect of the decision strategy.

2.5 Project Plan

Phase	Major Activities	Output
1. Domain study	Study multimodal hallucination, SHROOM/Mu-SHROOM, and attribute binding.	Problem definition and literature review
2. Dataset analysis	Inspect AMBER questions, identify paired attribute/state/action examples, and verify clean pairs.	Evaluation subset and parsing pipeline
3. System implementation	Implement OWLv2 grounding, cropping, SigLIP scoring, thresholding, and forced choice.	Working verification pipeline
4. Experimental design	Freeze development thresholds, construct four 2×2 cells, and define held-out test protocol.	Controlled ablation setup
5. Evaluation	Compute accuracy, bootstrap confidence intervals, controls, and artifact checks.	Experimental results
6. Reporting	Analyse findings, limitations, novelty, and prepare project report/presentation.	Final documentation

Table 3. Project implementation and evaluation plan.

3. TECHNICAL SPECIFICATION

3.1 Requirements

3.1.1 Functional

- Load the AMBER dataset and identify attribute-based questions and their candidate alternatives.
- Parse the target object and attribute information required for visual verification.
- Use OWLv2 for open-vocabulary detection of the target object.
- Generate a padded crop of the detected region while retaining the original image as a control.
- Use SigLIP to compute image–text similarity for competing attribute descriptions.
- Implement both independent threshold verification and comparative forced-choice verification.
- Run all four combinations of image representation and decision strategy on the same questions.
- Compute accuracy and image-level bootstrap confidence intervals on the held-out test set.
- Record detector fallback cases and maintain reproducible experiment configurations.

3.1.2 Non-Functional

- Reproducibility: thresholds and model revisions must be fixed before held-out testing.
- Reliability: failed object detections must fall back to the whole image and be explicitly reported.
- Efficiency: the system should operate within the memory limits of a student laptop GPU.
- Offline operation: inference should not require a paid API or external language model.
- Maintainability: the pipeline should be modular so that grounding, scoring, and decision rules can be tested independently.
- Fair evaluation: bootstrap resampling should occur by image because multiple questions can originate from the same image.

3.2 Feasibility Study

3.2.1 Technical Feasibility

The project is technically feasible because OWLv2 and SigLIP are open-weight vision models that can perform the required localisation and image–text scoring without task-specific training. The target laptop uses an RTX 3050 GPU with 4 GB of VRAM. By loading the models sequentially rather than simultaneously, peak VRAM usage is approximately 1.65 GiB.

3.2.2 Economic Feasibility

The system has zero monetary API cost. It does not depend on GPT-4V or another paid hosted model. Both models are open-weight and the implementation can be executed locally, reducing recurring inference cost and making the approach suitable for academic experimentation.

3.2.3 Social Feasibility

The project supports transparent and reproducible research by avoiding dependence on proprietary model outputs. The main ethical considerations concern benchmark validity, reporting of limitations, and avoiding claims that exceed the evidence. In particular, the project reports the marginal and non-replicated nature of the region-grounding effect rather than presenting it as a definitive mechanism.

3.3 System Specification

3.3.1 Hardware Specification

Component	Specification
Processor	Laptop-class CPU suitable for Python and model inference
Memory (RAM)	Sufficient system memory for AMBER preprocessing and model execution
Storage	Local storage for dataset, model weights, experiment outputs, and code
GPU	NVIDIA RTX 3050, 4 GB VRAM
Peak VRAM	Approximately 1.65 GiB during two-model staged execution

3.3.2 Software Specification

Software / Library	Role
Python 3.11	Primary implementation language
PyTorch	Model inference and tensor computation
OWLv2	Open-vocabulary object localisation
SigLIP	Vision-language image–text scoring
spaCy	Text/grammar processing where required for object and attribute parsing
NumPy / SciPy	Numerical processing and statistical calculations
pytest	Automated testing
Git	Version control and experiment traceability
AMBER	Primary evaluation benchmark

4. DESIGN APPROACH AND DETAILS

4.1 System Architecture

The proposed framework treats attribute hallucination detection as a verification pipeline. AMBER supplies an image and an attribute query. The target object is identified with OWLv2, a padded crop is produced, and both the full image and crop are retained so that region grounding can be evaluated as an independent experimental factor. SigLIP scores the candidate attribute descriptions. The decision module then applies either a threshold to an individual score or a forced-choice comparison between competing attributes.

Framework / Architecture / Block Diagram

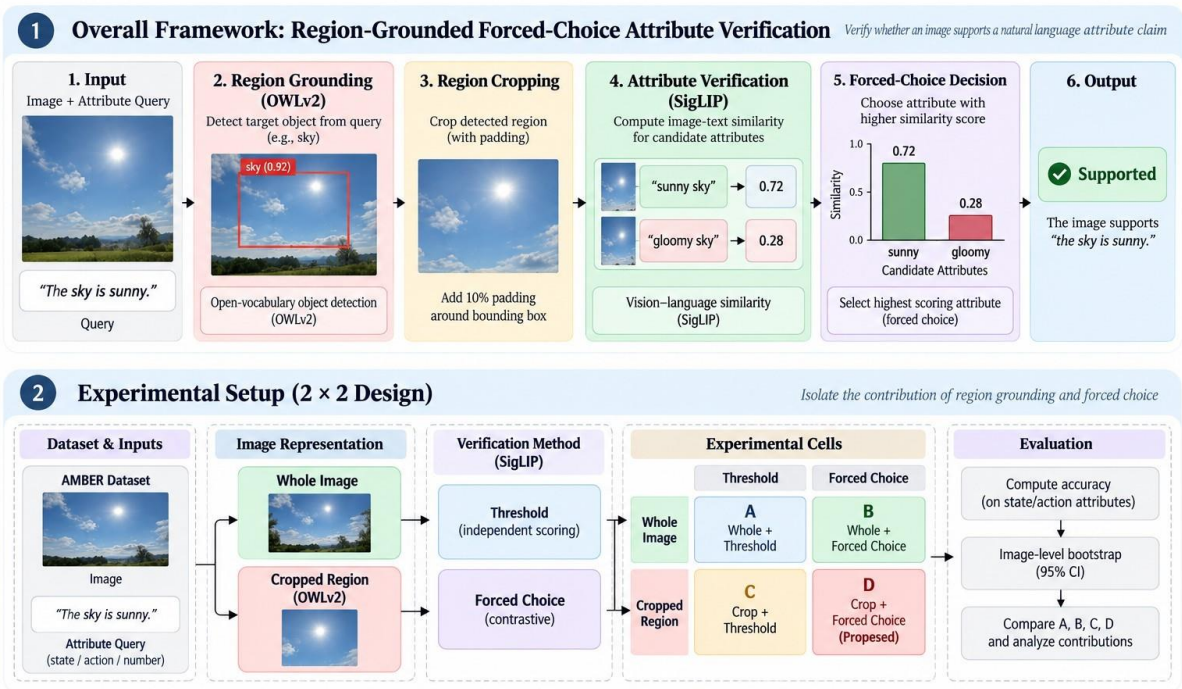


Fig. 1. Region-grounded forced-choice verification framework and controlled 2×2 experimental setup.

4.2 Design

4.2.1 Data Flow Description

1. AMBER provides the image, object, and attribute question.
2. The target object is detected with the prompt "a photo of a {object}" using OWLv2.
3. The highest-scoring surviving detection is expanded with 10% padding, clipped to the image, and retained as the cropped representation.
4. SigLIP evaluates "a photo of a {attribute} {object}" for each candidate attribute on both the whole image and crop.
5. The threshold strategy evaluates candidates independently against a fitted cutoff, while forced choice selects the higher-scoring candidate.

6. The predicted answer is compared with AMBER ground truth and accumulated for each experimental cell.
7. Image-level bootstrap resampling produces confidence intervals and paired comparisons between cells.

4.2.2 Use Case Description

Actor	Use Case	Description
Researcher	Prepare dataset	Load AMBER and construct clean attribute pairs.
Researcher	Run grounding	Detect the object named by the query using OWLv2.
Researcher	Run verification	Score candidate attributes using SigLIP.
Researcher	Select decision rule	Run threshold or forced-choice inference.
Researcher	Evaluate	Compute accuracy, confidence intervals, and ablation effects.
Researcher	Analyse robustness	Run base-rate and external-contrast controls and benchmark artifact checks.

4.3 Stepwise Implementation

4.3.1 Data Preparation

The AMBER dataset contains 1,004 photographs and approximately 14,216 discriminative queries across existence, attribute, and relation settings. The project focuses on attribute/state/action questions. Questions are parsed into the target object and attribute information. The evaluation path uses 2,378 clean state pairs and 396 action pairs, where the same image and object have one true and one false attribute. The paired structure supplies the competing alternatives for forced choice.

4.3.2 Region Grounding with OWLv2

OWLv2-base is prompted with “a photo of a {object}”. Detections with score at least 0.10 are retained, duplicate detections are removed using non-maximum suppression at IoU 0.5, and the highest-scoring surviving box is selected. The selected box is expanded by 10% on each side, clipped to the image boundaries, and constrained to a minimum size of 64×64 pixels. If no detection survives, the system uses the whole image and records a fallback flag. The observed fallback rate is approximately 3.4% of questions.

4.3.3 Region Cropping

The padded OWLv2 bounding box is cropped from the original image. The original full image is always retained. This is important because the full image forms the control condition in the 2×2 experiment; the crop is not substituted for the original image in every run.

4.3.4 Attribute Verification with SigLIP

SigLIP-base-patch16-224 is used as the vision-language scoring model. For each candidate attribute, the prompt follows the template “a photo of a {attribute} {object}”. The model returns a similarity score. The same scoring process is applied to both the full image and the cropped region.

4.3.5 Decision Strategies

Threshold-based verification evaluates an individual candidate score against a threshold τ . Thresholds are fitted on development data. Forced-choice verification compares the scores of the two candidate attributes and selects the higher-scoring alternative, requiring no free threshold parameter. Candidate order is fixed alphabetically so that option position cannot encode the answer.

4.3.6 Controlled 2×2 Experiment

	Threshold	Forced Choice
Whole Image	A — baseline	B — whole + forced choice
Cropped Region	C — crop + threshold	D — proposed

Table 4. Controlled 2×2 ablation design.

4.3.7 Evaluation Protocol

The development and test splits are separated by image so that no image occurs in both splits. Thresholds are frozen from development before the held-out test evaluation. Accuracy is computed over individual question IDs for comparability with AMBER reporting. Confidence intervals use 10,000 bootstrap resamples over images, and cell differences use paired image-level bootstrap comparisons because multiple questions from a single image are correlated.

4.4 Experiments and Results

Input	Threshold	Forced Choice
Whole Image	57.51%	77.87%
Cropped Region	59.53%	80.12%

Table 5. Held-out AMBER test accuracy.

The proposed cropped-region forced-choice configuration reaches 80.12% accuracy, compared with 57.51% for the whole-image threshold baseline. The overall improvement is

+22.60 percentage points. More importantly, the 2×2 design isolates the contributions of the two changes.

Comparison	Change	Test improvement
A $\rightarrow$ B	Forced choice on whole image	+20.36 percentage points
A $\rightarrow$ C	Region grounding with threshold	+2.01 percentage points
B $\rightarrow$ D	Region grounding after forced choice	+2.25 percentage points
A $\rightarrow$ D	Both changes together	+22.60 percentage points

Table 6. Attribution of the two experimental factors.

The central result is that the decision formulation contributes substantially more than cropping. Forced choice accounts for roughly ten times the improvement attributable to region grounding in the baseline contrast. The 77.87% versus 80.12% comparison shows that cropping adds only a small incremental gain after forced choice. The development-set interaction did not replicate on held-out test data and is therefore not treated as a confirmed effect.

4.4.1 Robustness and Fairness Controls

A base-rate-matched threshold control was used to address the possibility that forced choice benefits simply from the paired answer distribution. Matching the baseline’s global YES rate does not remove the forced-choice advantage. A second control replaces the benchmark-provided competitor with a fixed external antonym list committed before evaluation. The external-contrast variant achieves 76.02% on the matched covered test subset, a +19.01 percentage-point improvement over the corresponding baseline, showing that the gain does not depend entirely on the benchmark supplying the competing word.

The study also identified three AMBER construction artifacts that can allow image-independent strategies to achieve perfect accuracy in certain question types. These artifacts are reported rather than exploited. The state, action, and number question types used to support the main attribute result are free of the identified degenerate patterns.

4.5 Novelty and Contribution

1. Decision formulation: Instead of asking whether a single image–text score exceeds a cutoff, the proposed method asks which of two competing attribute descriptions scores higher. This removes the need for absolute score calibration at decision time.
2. Controlled isolation of components: Region grounding and forced choice are tested separately in all four combinations of a 2×2 experiment. This allows the contribution of each ingredient to be measured rather than reporting only a bundled improvement.
3. Lightweight, offline implementation: The system uses OWLv2 and SigLIP, totaling 358 million parameters, with no large language model, no paid API, and peak VRAM of approximately 1.65 GiB. This makes the method reproducible on a student laptop.

An important part of the contribution is the negative result: the project began with the expectation that cropping would be the key improvement, but the controlled experiment indicates that changing the decision rule is much more influential. The report therefore avoids presenting region grounding as the primary mechanism.

4.6 Limitations

- The current evaluation is English-only and uses AMBER’s closed object vocabulary.
- Forced choice requires a meaningful competing hypothesis; some action attributes do not have a natural English opposite.
- Approximately 3.4% of questions fall back to the whole image because OWLv2 does not produce a surviving detection.
- The effective sample is smaller than the raw question count because multiple questions share the same image and object/attribute structures.
- The region-grounding effect is small and its post-forced-choice test interval includes zero; further replication is required.
- The system is zero-shot and does not include training or calibration of the underlying vision-language models.
- Confidences produced by the vision-language scorer are not claimed to be calibrated probabilities.

4.7 Conclusion

This project developed and evaluated a region-grounded forced-choice framework for attribute verification in multimodal hallucination detection. The controlled 2×2 experiment shows that forced-choice decision making produces the dominant improvement, increasing whole-image performance from 57.51% to 77.87%, while region grounding alone increases threshold performance from 57.51% to 59.53%. The combined configuration reaches 80.12% on the held-out AMBER test set, a +22.60 percentage-point improvement over the baseline. The evidence therefore supports a narrow practical conclusion: when adding attribute verification to a multimodal hallucination pipeline, comparing competing hypotheses can be more valuable than relying on an absolute similarity threshold. Region grounding remains plausible as a small supporting component, but its independent effect is limited and requires further replication.

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